

Enhancing Autonomous Vehicle Safety with Fuzzy Logic-Based Collision Control in Traffic Lane

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Abstract—The quick development of autonomous car technologies calls for advanced methods to ensure road safety and efficient navigation. This work proposes the implementation of a traffic lane-oriented fuzzy logic-based collision control system. Fuzzy logic is the best approach for managing complex traffic dynamics since it can solve imprecision and uncertainty in practical situations. To perform informed, immediate judgments for accident avoidance, the designed controller evaluates variables like vehicle speed, distance to obstacles, and lane alignment. Simulation results show how well this technology improves safety, reduces collision risks, and helps to enable more seamless lane transitions. The main objective of this work is to develop a fuzzy logic-based collision control system fit for autonomous cars negotiating traffic lanes. This study emphasizes the need for fuzzy logic as a fundamental technology in the ongoing development of reliable and safe autonomous autos. The performance is measured in terms of velocity of vehicle and all simulations are done in MATLAB R2019a tool.

Keywords: *Autonomous Vehicle Safety, Fuzzy Logic, Traffic Lane.*

I. INTRODUCTION

Self-driving cars, sometimes referred to as autonomous vehicles, are those, which can navigate and run without human control. These vehicles make decisions on how to negotiate the road using a mix of sensors, cameras, and artificial intelligence to view their surrounds. Autonomous cars are meant to increase road safety, ease traffic congestion, and give people a more handy and quick way of mobility.

The concept of autonomous vehicles has been extensively useful in industry today. It helps to maximize safety and ensure driver convenience in terms of moving vehicles. In today's time, 95% of accidents are just due to human error. Many researchers

are doing work related to autonomous systems to reduce vehicle accidents. While regular cars come with some amenities to help prevent drivers from becoming sleepy, smart autonomous cars use wireless networks and smart software to lower the number of accidents on the road [1]. The reason for most of the accidents is either unknown or a mistake on the other side. Therefore, to handle these reasons, the autonomous concept is an emerging thing.

The use of autonomous driver assistance systems has become popular in preventing road injuries. These systems are required in India on an urgent basis as road accidents are increasing day by day. 80% of accidents are due to vehicle collisions or crashes for any reason. This condition is similar in all areas of the world. Many efforts are going on to improve vehicle advancement and smartness to enhance road safety.

Vehicular Ad hoc network (VANET) [2] is a network in which vehicles act as nodes, and they are all dynamic, so it is called vehicular ad-hoc network. This is also a good initiative to enhance safety on the roads. Any vehicle consists of a vehicle node, a device sensor, a roadside unit, and a traffic management center. Some vehicles are moving at a faster rate, which can cause concerns related to data delivery and contact crashes. Due to this, communication can affect latency performance. The newest technology is crucial for car safety because it is developing more quickly. It can provide solutions for the problems that occur due to conventional reasons. When a vehicle finds an obstacle in its path, it is necessary to change the path on an immediate basis. A conventional vehicle takes time and effort to do this task, but a smart vehicle with a smart system can immediately detect the alternate path without obstacles and give instructions with surrounding parameter detection to achieve the destination in less time. Similarly, in the case of multi-vehicle

movements, corner collision is an issue and one of the concerns for road accidents.

A. Importance of safety in autonomous vehicles

Gaining public confidence and support for this developing technology depends on ensuring safety in autonomous cars. Autonomous cars have the chance to drastically cut the number of accidents and deaths on the road since they could help to lower human error [3]. Manufacturers may help to reduce risks and provide a safer transportation system for all by giving safety measures including strict testing, cybersecurity policies top priority. Fail-safe features also help to lower risk. Eventually, the effective deployment of autonomous cars depends on building a robust basis of safety measures to guard pedestrians as well as passengers. This basis will not only inspire faith in technology but also open the path for general acceptance and integration of driverless cars into our daily life. Safety must always come first even as society develops and adopts new technologies to guarantee a seamless shift to a more automated future. Working together to solve issues and always enhance safety precautions will help us to fully realize autonomous cars and transform our journey.

II. REVIEW OF LITERATURE

An innovative Wi-Fi-based pedestrian collision avoidance (PCA) system called Wi-Safe was being introduced [4]. One major advantage of Wi-Safe over vision-based and radar-based PCA systems is its non-line-of-sight (NLOS) capability. When a structure, a car, or another obstruction blocks a driver's view, it creates a blind spot. Wi-Safe can help drivers find pedestrians in these situations. Split seconds can be saved using Wi-Safe, potentially saving lives. Wi-Safe offers a cheap, easy, and practical alternative to WAVE/DSRC for pedestrian protection. This publication gave the whole analysis, implementation, and experiments. The outcomes demonstrate that Wi-Safe can meet and even surpass the PCA specifications.

Class EF and Class E/F inverters are hybrid inverters that combine the efficient switching of Class E inverters with the better switch voltage and current waveforms of Class F and Class F-1 inverters [5]. There was little information available concerning these inverters, and no attempt was made to offer a thorough evaluation of their functionality. Following a more thorough investigation, three unique operation scenarios for the Class EF2 inverter were established, enabling it to function at its maximum power-output capability. To facilitate quick design and development, final design equations were supplied. Additionally, the results demonstrate that the Class EF2 inverter's efficiency of 91% was attained in contrast to its Class E inverter's 88% efficiency.

Highly advanced and dependable driver assistance systems may malfunction if a driver is not completely aware of the features and capabilities of the system [6]. Two groups of twenty-four drivers each had to avoid multiple approaching dangers. Driver responses and behavior varied significantly depending on which assistance system was used in response to whatever danger. When both systems hit operational limits that the drivers misinterpreted, more collisions were seen. The effect of these restrictions, however, differed based on the driver's capacity to regain control and salvage the crucial circumstance.

The factors that influenced the subjective post-hazard assessments of drivers were the critical conditions and the ease with which control authority was transferred. This shows how important it is to have adaptive automation that can balance the system's and humans' processing capacities with the potentially dangerous situations that arise.

Path planning for autonomous underwater vehicles in complex, unfamiliar maritime environments is still a difficult task. To address the issue, a novel path-planning technique based on the prediction of motion parameters for moving objects (OMPP) was presented [7]. To extract distinct objects from sonar data, a fast random search tree approach was created. The prediction of object motion parameters was replaced with a virtual centroid. A cooperative autonomous driving system based on collective perception was proposed in [8]. The suggested cyber-physical system-based heuristic approach may be successfully implemented in multiple autonomous guided vehicle logistics systems to further enhance system security and operational effectiveness, according to the results of the numerical experiments.

Trajectory planning is a frequently used technique to automatically produce an expected geometric path [9]. It is assumed that every vehicle communicates with the server (such as roadside devices or cloud servers) in its intended direction as well as its current local state (such as position or speed). After gathering the data, the computer calculates if there will be any collisions among the group of cars in a predetermined amount of time. To prevent collisions, the server then computes the altered trajectories and sends them back to the cars. Calculating such trajectories was often a hard job. It offered two ways to shorten the time it takes to compute the trajectories in a driving scenario with several cars engaged. Maritime autonomous surface ships (MASS) must be sufficiently safe to be accepted commercially [10]. Nevertheless, current collision risk assessment techniques do not consider all the necessary elements for collision avoidance that comply with COLREG regulations. It suggested an inference system for collision risk for MASS that adheres to the essential guidelines set forth by COLREG for collision avoidance. Unlike previous studies, the suggested method improved the timing and positioning of the possible collision warning by considering all relevant factors in the COLREG rules and collision avoidance guidelines. As a result, it might give decision-makers more time to take the required precautions against collisions.

The Vehicular Sensor Network (VSN) has received a lot of academic attention because of its enormous potential for environmental monitoring [11]. However, it remained challenging to reduce data collisions and broadcast storms in the context of vehicle sensors. Collisions occur because of incorrect safety message broadcasting and simultaneous packet delivery from several vehicles. Our main goal was to overcome these VSN weaknesses. As a result, it suggested using cluster-based VSNs for novel segment-based safety message broadcasting. In this case, the best forwarder was chosen to transmit a safety message and lessen the broadcast storm.

An autonomous vehicle's architecture must have mechanisms for handling critical scenarios [12]. A cautious approach that assumes the least amount of tyre-road friction for which the

maneuver is still possibly served as the foundation for the development of an autonomous collision avoidance controller. If there is more friction, the controller takes advantage of it and operates more effectively. The controller uses an acceleration-vector reference that was found using friction-limited particle optimum control. It was shown to work by numerically optimizing a model of the whole vehicle. This led to the creation of a controller whose output computation was explicit and unaffected by the tyre-road friction. The controller created performs near that of offline numerical optimization when tested in real-time on a high-fidelity simulation model [13].

By means of fuzzy logic in collision control, autonomous cars can make split-second decisions in challenging and uncertain environments, therefore enabling their ability to lower the accident risk. Autonomous cars may examine and react to many elements in real-time by including fuzzy logic, therefore guaranteeing a better and safer driving environment for all road users. Using fuzzy logic in autonomous cars offers generally better safety, efficiency, and a smoother integration of self-driving technologies into current transportation systems.

III. RESEARCH METHODOLOGY

This work provides the concept of fuzzy inference system controller to control vehicle's movement. It integrates the learning potential of networks with the explicability of fuzzy logic. The main objective of this work is to develop a fuzzy logic-based collision control system fit for autonomous cars negotiating traffic lanes. This technique facilitates exact regulation of a vehicle's motion, considering diverse environmental variables and input data. This work examines the elements of a fuzzy controller and its utilization in autonomous vehicles. Analyze case studies and scenarios provided to demonstrate the efficacy of fuzzy controllers in autonomous cars. This work seeks to elucidate the function of fuzzy controllers in the progression of autonomous vehicle technology.

More complicated decision-making in uncertain events made possible by fuzzy logic helps vehicles negotiate difficult traffic conditions with more accuracy. Implementing fuzzy logic-based collision control systems helps us to reduce the uncertainty and human unpredictability, therefore improving the safety of our roadways for all. Autonomous cars made possible by fuzzy logic technology may make decisions depending on a range of criteria, including weather, visibility, and the conduct of other drivers, and adjust to changing road conditions. By rapidly evaluating possible collision hazards and acting appropriately for evasive action, this sophisticated technology can help avoid accidents. Using fuzzy logic in collision management helps autonomous cars run more securely and effectively, therefore lowering the possibility of collisions and enhancing general traffic flow.

A. Modelling Obstacle Detection System

The designed fuzzy logic controller (FLC) for the navigation task employed two inputs: angle orientation (φ) and distance (d). The left wheel's and right wheel's speeds are the controller's outputs. For a variety of applications where distance estimates are essential, the sensor offers precise measurements. The method for the distance calculation from an obstacle:

$$d = \frac{\text{time of flight}}{2} * \text{velocity of sound } (c) \quad (1)$$

The walking velocity of pedestrians is assumed to be zero in the purpose of this research and is ignored with respect to automobile A. Thus, it is possible to calculate the critical braking distance between vehicle A and the pedestrian.

$$d_{c1} = V_A \left(t_r + \frac{t_i}{2} \right) + \frac{V_A^2}{2\mu g} + d_{min} \quad (2)$$

$$d_{c2} = V_A t_r + \frac{(V_A - V_B)t_i}{2} + \frac{V_A^2 - V_B^2}{2\mu g} + d_{min} \quad (3)$$

Where d_{c1} & d_{c2} are the braking distances for cars A and B respectively.

B. Creating Fuzzy Rules for the First Fuzzy Logic Controller

Mamdani's fuzzy logic approach is applied in this study. Two elements affect the value of the automobile prototype's brakes in this fuzzy logic system. The first is the car's distance from the barrier in front of it. The closer the vehicle gets to the obstruction, the harder it must brake to prevent a crash. The second factor is the vehicle's speed; if the vehicle is travelling at a high speed, it will require more braking, and if it is travelling at a low speed, the fuzzy logic system will produce less braking. Two input variables, distance and speed that will be used in the fuzzy logic system are derived from the primary factors, and the system's output variable is the braking angle. Moreover, the creation of input and output variable membership functions plays a crucial part in how the problem is presented as shown in Fig.

These are primarily Inputs for Navigation Task: Angle Orientation (θ) and Distance (d) Outputs of the Controller: Left Wheel and Right Wheel Speed Sensor Capabilities: Precise Distance Measurements Important Factors to Consider: Maintain Minimum Distance (d) between vehicles Normal Operating Conditions: Maintaining Present Speed Automated Safety Response: Instant Activation of Automated Brake Slowdown Collision Avoidance with another car

Autonomous cars can make more wise lane changes considering several factors including speed, distance between cars, and road conditions. This can help to prevent mishaps and lower the possibility of collisions, so rendering the roads safer for all. Further improving traffic lane safety is the inclusion of fuzzy logic into lane change algorithms, which helps autonomous cars to adjust to shifting traffic conditions and unanticipated obstructions. Autonomous cars fitted with fuzzy logic can negotiate traffic more successfully and efficiently by always assessing and modifying their lane change actions depending on real-time data. Smoother lane changes made possible by this modern technology help to lower the possibility of sudden maneuvers that could frighten other drivers. All things considered, including fuzzy logic into lane change decisions could transform road safety and open the path for a time when traffic-related injuries and accidents are infrequent.

C. Details of Activation Phase

In this stage, it determines the activation region based on the vehicle specs and VRU. The critical factor in determining this area is the distance at which the driver detects a potential hazard

and initiates braking to bring the vehicle to a halt. The stopping distance is defined as:

$$d_{stop} = \frac{v_v^2}{2g(\mu+G)} \quad (4)$$

The variables in the equation are as follows: v represents speed of the vehicle, G represents percentage roadway slope, μ represents friction coefficient that varies depending on road surface, and g represents acceleration of gravity, which is equal to 9.8 m/s². Furthermore, it calculates upper and lower longitudinal distances between Vulnerable Road Users (VRUs) and cars using Equations (5) and (6). There are two purposes of vehicle speed at these distances.

$$d_{front}(min) = (t_{reaction}(min) \cdot v_v) + d_{stop} \quad (5)$$

$$d_{front}(max) = (t_{reaction}(max) + TTC_v(critical)) \cdot v_v \quad (6)$$

The term $t_{reaction}(min)$ refers to the minimum time required for motorists to react after recognizing danger, which is typically 1 second. Essentially, lower limits refer to the minimum amount of time motorists must prevent accidents after hearing an alarm. Similarly, maximum driver reaction time, often known as reaction (max), is three seconds. The TTC_v (critical) is precisely three seconds. Finally, the velocity of the vehicle is represented by the variable v_v . These parameter settings resulted in a mean brake reaction time of 1.5 seconds. In the most unfavorable scenario, this time might potentially increase to 2.5 seconds, depending on perception-reaction time.

Autonomous cars' fuzzy logic-based collision control systems present a potential fix to increase safety and lower road accident risk. These systems can efficiently negotiate difficult traffic conditions and prioritize the safety of all road users by using fuzzy logic to make real-time judgments depending on ambiguous or imprecise inputs as shown in Fig1. Further improving their efficacy in averting collisions is the adaptive character of fuzzy logic, which lets these systems learn and adjust to changing surroundings. By means of fuzzy logic-based collision control systems, autonomous cars may better predict and react to possible threats, therefore enabling a safer and more effective transportation network for everybody. Even in erratic circumstances, fuzzy logic-based collision control systems can enable autonomous vehicles to make split-second decisions to avoid accidents by always assessing and reacting to real-time input. This capacity to react fast to possible hazards not only increases passenger and pedestrian safety but also helps traffic flow generally and lessens road congestion. The future of transportation will surely be shaped in great part by the inclusion of fuzzy logic into autonomous cars as technological developments keep changing.

Apart from collision control, fuzzy logic technology is absolutely important for improving the general performance of autonomous cars. These vehicles can negotiate challenging traffic situations with accuracy and simplicity by always assessing and modifying many factors, including speed, acceleration, and braking. Their great degree of adaptability lets them react fast to abrupt environmental changes, so guaranteeing a flawless driving experience for the passengers. Moreover,

fuzzy logic helps maximize fuel economy and lower pollutants, thus autonomous cars not only are safer but also more ecologically beneficial. All things considered, the incorporation of fuzzy logic technology in autonomous cars has significant potential to transform travel and help to define the direction of transportation. A self-driving car with fuzzy logic technology, for instance, can modify its speed and braking to identify and react to erratic road conditions including unexpected heavy rain or snow. This flexibility guarantees that the car may negotiate difficult surroundings without endangering passenger comfort or safety.

A primary advantage of employing fuzzy controllers in vehicle management is their capacity to adapt and learn from real-time data, facilitating more efficient and accurate operation of autonomous cars. During the implementation process, meticulous attention must be paid to the compatibility of fuzzy controllers with current vehicle systems, along with any requisite modifications for seamless integration. By confronting these difficulties directly and suggesting novel solutions, it optimizes the capabilities of fuzzy controllers to transform the domain of autonomous vehicle technology.

D. Details of Warning Phase

This suggested strategy considers vehicles' Time to Collision (TTC), acceleration, speed, and differentiation between pedestrian and vehicle TTC variables. In this scenario, it additionally considers the impacts of environmental elements as well as the characteristics that affect drivers and pedestrians.

The membership function for fuzzy set $A \sim$, as depicted in figure 2, represents the precise region within the universe where there is complete and absolute membership in set $A \sim$. The core comprises the elements x in universe for which the membership function $\mu_{A \sim}(x)$ equals 1. The support of membership function for a fuzzy set $A \sim$ is portion of universe where the membership in the set $A \sim$ is non-zero. The support comprises elements x in universe for which degree of membership $\mu_{A \sim}(x)$ is larger than 0. The left wheel velocity and right wheel velocity are defined in terms of membership functions for the linguistic variables fast, medium, and slow. The membership functions for the left wheel velocity and right wheel velocity incorporate linguistic phrases such as very slow, slow, medium, fast, and very fast. The fundamental details of fuzzy controller is shown in Table 1 below.

TABLE I. FUNCTIONAL DETAILS OF FUZZY CONTROLLER (SOURCE: MATLAB)

Membership Function	ZMF (Z-shaped membership function)
Defuzzification	Weighted average
And Method	Product
Or Method	Probabilistic OR (also known as the algebraic sum)

In the context of fuzzy sets, when there is exactly one element with membership value of one, it is commonly known as the prototype or prototypical element of the set. The process of generating rules for fuzzy inference systems is frequently the most challenging aspect of design process. Typically, a certain level of expertise in understanding the dynamics of the plant is necessary.

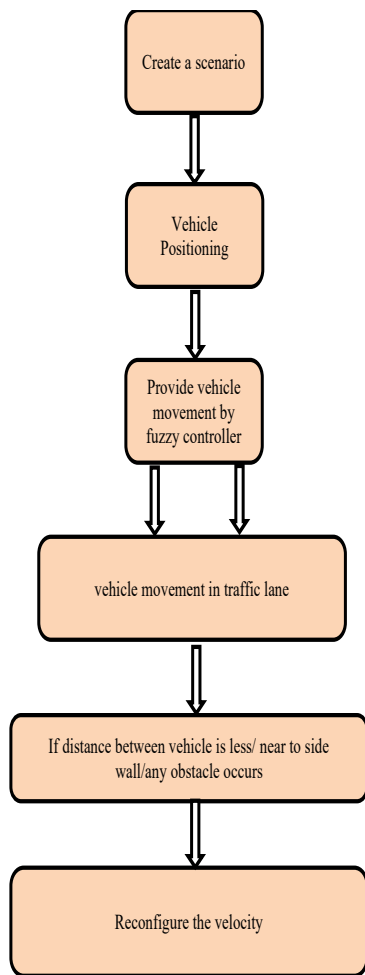


Fig. 1. Designed Architecture of System

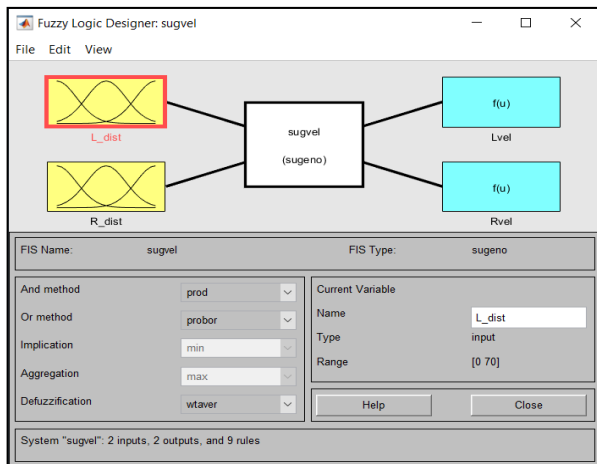


Fig. 2. Fuzzy Controller using Sugeno Model (Source: MATLAB)

IV. RESULTS OF THE SYSTEM

While fuzzy logic systems can adjust and react in real-time to changing road circumstances, traditional collision control techniques sometimes depend on set rules and reactions. Real-world testing has revealed encouraging outcomes from this adaptability and efficiency, therefore confirming the possibility

of fuzzy logic technology in transforming traffic control and road safety.

Researchers at a big automaton firm found in one study that vehicles with fuzzy logic collision control systems might cut the accident when compared to those running conventional systems. Along with saving lives, this notable drop in crashes lessened the general cost of accidents for insurance companies and drivers. Further providing drivers with further road protection was fuzzy logic technology's ability to forecast and stop possible collisions before they ever happened. These outstanding achievements have attracted attention in using fuzzy logic systems in a variety of vehicles, from buses and even autonomous cars to cars and trucks.

The results are shown based on the scenario for Vehicle moving in traffic area. The parameters of velocity and distance are presented using a fuzzy controller. Four vehicles are used in this scenario to check the effectiveness of the system. On the graph, a horizontal line denotes a constant speed, therefore guiding the vehicle toward uniform speed. Usually, this phase results from resistive forces balancing the driving force and reaching equilibrium. Somewhere in the graph, a growing trend indicates probably the acceleration phase of the vehicle. External forces like engine power overcome resistance (e.g., friction and drag) during this period thereby enabling the vehicle to acquire speed. The graph shows erratic fluctuations that would point to changes in driving conditions including traffic, roadblocks, or varying engine performance. Many autonomous car prototypes now under testing on public roadways include fuzzy logic-based collision control systems. These technologies have demonstrated encouraging results in lowering the number of accidents resulting from human mistakes, therefore opening the path for a time when autonomous cars are the standard. Furthermore, the incorporation of fuzzy logic in collision control has shown to greatly reduce traffic congestion and enhance general traffic flow, therefore enabling a more effective and stress-free driving environment for all road users. Moreover, the real-time decision-making capacity of fuzzy logic systems in collision control has shown their capacity to adjust to shifting road conditions and unanticipated roadblocks, therefore improving the safety of autonomous cars even more. The probability of general acceptance of autonomous vehicles gets more realistic as more study and development are focused on refining these systems. The developments in fuzzy logic technology are fundamentally changing our perspective on transportation and are likely to have a major influence on traffic efficiency and road safety in not-too-distant future.

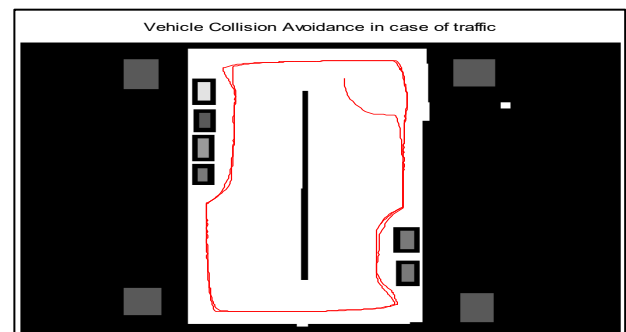


Fig. 3. Path Traced with Four Vehicles in Traffic Lane/Obstacle

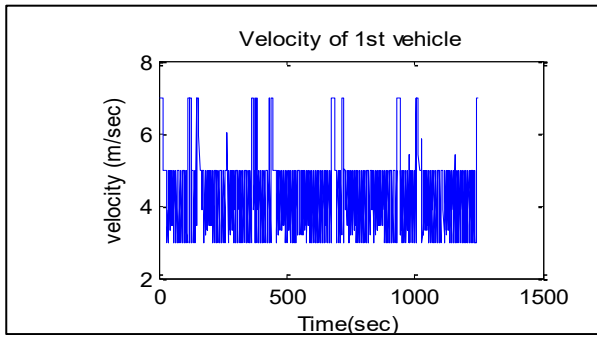


Fig. 4. Velocity of First Vehicle in Traffic Lane

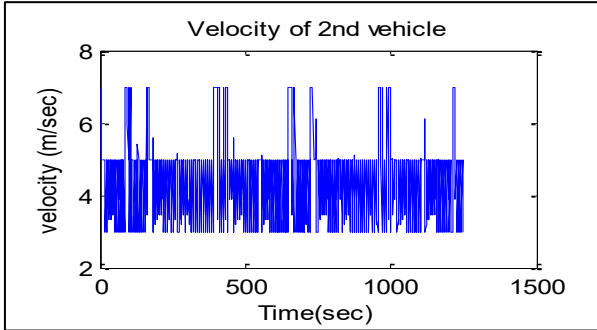


Fig. 5. Velocity of second Vehicle in Traffic Lane

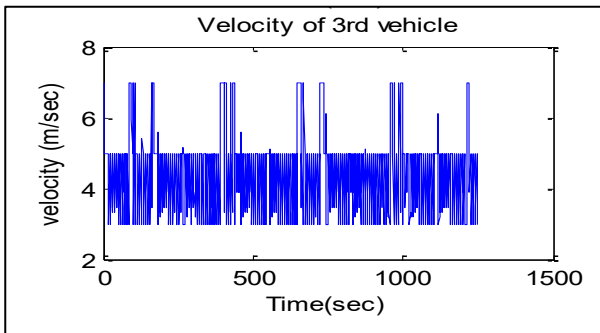


Fig. 6. Velocity of third Vehicle in Traffic Lane

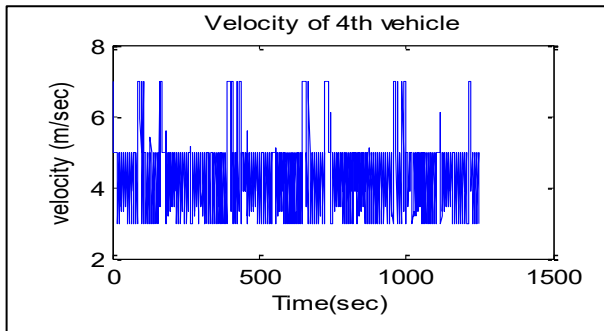


Fig. 7. Velocity of fourth Vehicle in Traffic Lane

V. CONCLUSION AND FUTURE IMPLICATIONS

The inclusion of fuzzy logic-based collision control in autonomous cars marks a major progress in guaranteeing traffic lane safety. Through efficient management of imprecise inputs and uncertainty, the fuzzy logic controller provides a strong means of navigating, challenging and changing traffic

conditions. The suggested method shows how well it could increase real-time decision-making efficiency, vehicle safety, and collision risk lowering ability. This method not only advances autonomous systems technologically but also fits the general goal of building more dependable and safer transportation networks. The paper emphasizes the important part fuzzy logic plays in developing intelligent cars and provides groundwork for next investigation on flexible and effective collision control systems.

- Effective integration of fuzzy logic-based collision control could open the path for intelligent and more flexible autonomous car systems able to manage challenging real-world traffic situations.
- Adoption of such technology could greatly lower traffic accidents, therefore providing safer surroundings for pedestrians, passengers, and drivers.

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